

Help Document for PyWave FDTD GUI

Running Project:

- Ensure you are logged into ece-acemspi-02.rdp.uidaho.edu
 - If using ssh, make sure to enable remote window popups with
`ssh -X username@ece-acemspi-02.rdp.uidaho.edu`
- Navigate to the SrDesign_PyWave_F2025_S2026 directory
- Enter dependency safe environment by running `poetry shell`
- Run program with `python3 PyQt5-test/input-screen/main.py`

Demos:

All demo files created and used in videos are saved in the demos folder.

- Demo 1: https://youtu.be/GA7z-kR5_Uk
 - 2DTE wave impedance with one waveguide
 - Shows running simulation from the terminal
 - Longer video going into many of the GUI settings
- Demo 2: https://youtu.be/1lyCq_o03E0
 - 2DTM wave impedance with multiple waveguides and sources
- Demo 3: <https://youtu.be/1WU8xoqhD1A>
 - 2DTE scattering loss with just one source, then with a waveguide
 - Also shows restarting application and archiving projects